

Julia V.E. Kim

✉ julia225@mit.edu ☎ (+44) 7570 676922 🌐 juliavekim in juliavekim

Education

Trinity College, University of Cambridge

MASTERS OF MATHEMATICS

- **Scholarship:** Trinity Studentship in Mathematics

Cambridge, UK

Oct. 2024 - Jun. 2025 (expected)

University of Toronto

HONOURS BACHELORS OF SCIENCE IN PHYSICS, MATHEMATICS AND STATISTICS

- **Cumulative GPA: 4.00/4.00, Average: 97.6%, Rank: 1/16,000**
- **Scholarship:** Rose Sheinin Award for *top female student* in science at UofT

Toronto, CA

Sep. 2019-Jun.2024

Professional Experience

Data Science Intern

THE TRADE DESK: AI LAB

London, UK

Jun. 2025 - Aug. 2025

Astrophysics Machine Learning Researcher

UNIVERSITY OF TORONTO

Toronto, CA

May 2024-present

- Designed and trained deep convolutional neural networks in **PyTorch** to predict stellar parameters with high precision and accuracy
- Developed self-supervised learning models, successfully embedding cross-modal data into shared latent space
- Achieved an R^2 accuracy of over 95% on one-shot stellar property estimation tasks, outperforming traditional baselines by 19%

Biophysics Machine Learning Researcher

UNIVERSITY OF TORONTO

Toronto, CA

May 2023-May 2024

- Built a stochastic model in **TensorFlow** to simulate receptor signalling pathways with cross-talk and pleiotropy
- Applied information theory and statistical inference to investigate the impact of feedback and molecular noise on signalling specificity
- Improved model accuracy by 30% through hyperparameter tuning, feature selection and optimising neural network architecture

Dark Matter Particle Physics Researcher

SUPERCDS SNOLAB & UNIVERSITY OF TORONTO

Toronto, CA

Sep. 2021-Apr.2022

- Developed likelihood-based statistical models in **C++** to establish sensitivity projections for detecting dark matter signals
- Performed over 1,000 Monte Carlo simulations to generate test statistics for LIP detection at 95% confidence level
- Leveraged **Roofit** and **Roostats** to estimate an upper limit of 120 signal events, validating results with χ^2 and Gaussian distributions

Advanced Photovoltaics Researcher

AP²D LABORATORIES

Toronto, CA

Sep. 2021-Apr.2022

- Conducted pulse IV testing to uncover neuromorphic behaviors in novel memristor devices
- Assisted in photolithography lift-off processes and performed optical microscopy probing.
- Compiled and analysed data in **Python** and **JAVA** to evaluate device performance and neuromorphic properties.

Skills

Languages: Python, MATLAB, R, bash/command line, LaTeX

Libraries/Frameworks: PyTorch, TensorFlow, Numpy, Pandas, Matplotlib, Scikit-learn, Seaborn

Tools: Git, XCode, Microsoft Office Suite

Publications

1. Julia V.E. Kim & John R. Percy, "Pulsating Red Giants in a Globular Cluster: ω Centauri", 2022, JAAVSO, 50, 2. [Link](#) 🔗.
2. Natasha Leclerc, Julia V.E. Kim et al. 'Ba Depletion in Coastal Waters Points to Planktonic Phytoplankton Intensification'. (in progress)
3. Julia V.E. Kim. 'Patterns and Trends in Endangered Species Listing Heavily Biased Towards Imperiled Vertebrates with High Utility Value to Humans.' Featured in Alexander, R. (2024) *Telling Stories with Data with Applications in R*. Milton: CRC Press LLC. [Link](#). 🔗
4. Julia V.E. Kim. 'Significant Declines in Standardised Test Scores due to COVID-19 School Closures Disproportionately Affect Vulnerable Students.' Featured in Alexander, R. (2024) *Telling Stories with Data with Applications in R*. Milton: CRC Press LLC. [Link](#). 🔗

Projects

Using a Deep Deterministic Gradient Policy (DDPG) to Control a Two-Joint Arm

MACHINE LEARNING COURSE PROJECT

University of Toronto

Mar.-Apr. 2023

- Implemented DDPG reinforcement learning algorithm in **MATLAB** to optimise the motion of a two-joint robotic arm
- Developed and trained neural networks with ReLU activation, stochastic gradient descent and Adam optimisation

- Achieved improved policy returns over 1, 000 rollouts, by leveraging test policies and minibatch learning with replay buffer

Learning MNIST Handwritten Digits with a Retinotopic Map

MACHINE LEARNING COURSE PROJECT

University of Toronto

Feb.-Mar. 2023

- Implemented neural network to recognise digits, using backpropagation with Adam and Nesterov in MATLAB
- Developed 3-layer mapnet with ReLU activations, applying softmax for classification and training through minibatch learning
- Reduced classification error from 480 to below 170 incorrect guesses over 10 epochs across 60, 000 training and 10, 000 test examples

Awards & Honours

Elected to Sigma Xi, Sigma Xi	
University of Toronto Dean’s List Scholar, University of Toronto	2020-2024
Rose Sheinin Award, University of Toronto	Jun. 2024
St Micheal’s College Achievement Award in Maths and Physical Sciences, University of Toronto	Jun. 2024
Governor General Silver Medal Nominee, University of Toronto	Jun. 2024
University of Toronto Excellence Award, University of Toronto	Jun. 2023, Jun. 2024
Trinity Studentship in Mathematics, University of Cambridge	Apr. 2024
John J Sebisty Scholarship, University of Toronto	Mar. 2024
David and Louise Fraser Scholarship, University of Toronto	Mar. 2024
Hymie and Roslyn Mida Student Award in Theoretical Physics, University of Toronto	Mar. 2024
Lucia Colavita Memorial Scholarship, University of Toronto	Jun. 2023
Jackman Scholars-in-Residence Award, Victoria College	Jun. 2023
The 3T0 M.&.P. and Associates Scholarship, University of Toronto	Dec. 2021, Feb. 2023
Patricia Massel 5T1 Bursary, University of Toronto	Jun. 2022
NSERC USRA and University of Toronto Chair Scholar, NSERC	May 2022
St Micheal’s College In Course Scholarship, University of Toronto	Mar. 2022
University of Toronto In-Course Scholarship, University of Toronto	Sep. 2021
University of Toronto Scholar’s Award, University of Toronto	Aug. 2021
Samuel Beatty In-Course Scholarship, University of Toronto	Jan. 2021
Joseph Alfred Whealy In-course Scholarship, University of Toronto	Sep. 2020
Ontario Scholar & Governor’s General Academic Medal, Humberside Collegiate Institute	Jun. 2018
Certificat d’Honneur d’Études Bilingues en Français Immersion, Humberside Collegiate Institute	Jun. 2018
Diplôme National du Brevet des Collèges avec Mention Très Bien, Toronto French School	Jun. 2016

Teaching

University of Cambridge, Examinations Invigilator	Oct. 2024-present
Scholar Radiance, Advanced Mathematics Tutor	Jan. 2022-present
OCAD University, SCTM-2003: Astronomy–Scales & Structures TA	May-Jun. 2023
University of Toronto, MAT137Y1: Calculus! TA	Sep. 2020-May 2021
University of Toronto, MAT233: Linear Algebra I workbook editor	May-Jul. 2020

Talks

1. ‘Applying Self-Supervised Learning to the Analysis of Stellar Spectra’, SURP Presentation Day. University of Toronto (Aug. 2024).
2. ‘Ba-Ca and stable C isotope ratios in *C. compactum* suggest increase in Artic primary productivity driven by sea loss and higher nu-
trient input from permafrost melt and terrestrial run-off’, ASLO Conference, Madison, Wisconsin, USA (Jun. 2024).
3. ‘Replication Analysis of COVID-19 Test Scores’, DSI Reproducibility Challenge. Data Science Institute, University of Toronto (May 2024).
4. ‘Investigating *C. compactum* as an Artic Erosion Proxy’, UTM Indicium Talk. University of Toronto (Jul. 2023).
5. ‘Generalising M_{T2}^{DC} for Decay Chains and Mass Measurements’, ATLAS Talk. University of Toronto (Jul. 2022).

Outreach

Trinity College Cambridge, APPD Programme Assistant	Oct. 2024-present
Girls in STEAM, Vice President of Content	Jan. 2023-May 2024
University of Toronto Machine Intelligence Team, Academic Coordinator	Jan. 2023-May 2024
University of Toronto Aerospace Team, Structures Division Assistant	Jun. 2021-May 2023
Canada’s C3 CyberSTEAM Challenge, Director of Virtual Design Team	Sep. 2022-Jun. 2023
The Varsity, University of Toronto Newspaper Science Writer	Jun. 2022- Jun.2023

Media

Kaster, J. (2024) *St. Mike’s Julia Kim wins top U of T science award*, St. Micheal’s College, University of Toronto. [Link](#)